University of Calgary
Faculty of Science Midterm
PHYS 365 & PHYS 369

VERSION 01

October 29, 2024, in class Time: 75 minutes

This is a closed-book exam containing a total of 20 equally weighted questions.

ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the exam questions, the formula sheet, and 3 blank pages. This booklet will be handed in, but it WILL **NOT** BE GRADED.

2. A BUBBLE SHEET used to answer multiple-choice questions. This will be handed in and it WILL BE GRADED.

IMPORTANT: Start by entering your UCID, NAME, COURSE, LECTURE SECTION, and VERSION NUMBER on the top of the bubble sheet.

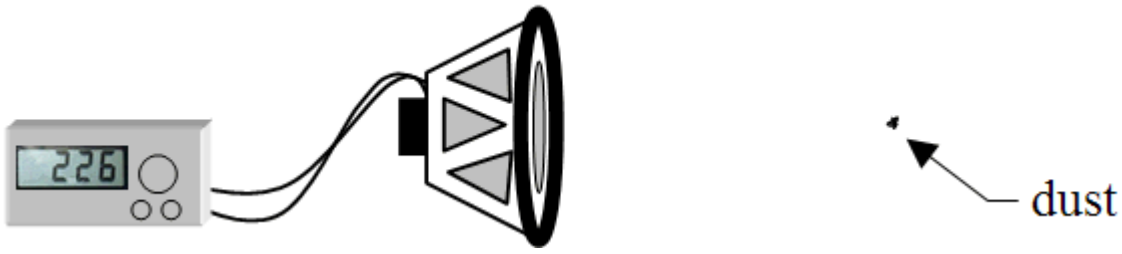
3. A CALCULATOR. Only a Schulich-approved scientific calculator may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet. If you are using a PEN, make sure it is BLACK or BLUE to ensure the scanner reads your answers properly. PENCILS can also be used, but make sure you press firmly when answering so the scanner reads your answer properly.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

6. Students remaining after **65 minutes** must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

1. A particle of dust is floating in the air approximately half a meter in front of a speaker, which produces a pure tone of 226 Hz, as shown in the figure below. The sound waves produced by the speaker travel horizontally. Which one of the following statements correctly describes the subsequent motion of the dust particle?



A) It will oscillate left and right with a frequency of 226 Hz.
B) It will oscillate up and down with a frequency of 226 Hz.
C) It will accelerate to the right and continue moving in that direction.
D) It will remain motionless as it cannot be affected by sound waves.

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2. A string is 2.70 m long, has a mass of 260 g, and is under a tension of 36.0 N. What must be the frequency of travelling waves of amplitude 7.70 mm for the average power to be 85.0 W?

A) 1241 Hz
B) 198 Hz
C) 1540 Hz
D) 343 Hz

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3. The following two waves travel along the same string (x is in meters and t is in seconds):

$$y_1 = (5.00\text{ mm})\sin(4.00\pi x - 7.00\pi t)$$
$$y_2 = (5.00\text{ mm})\sin(4.00\pi x - 7.00\pi t + 0.800\text{ rad})$$

What is the amplitude of the resultant wave $y = y_1 + y_2$?

A) 9.21 mm
B) 4.65 mm
C) 3.49 mm
D) 6.98 mm

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4. The tension of a guitar string is increased by a factor of 4. How does the wave speed on the string change after this increase?

   A) The wave speed increases by a factor of 2.
   B) The wave speed decreases by a factor of 1/4.
   C) The wave speed decreases by a factor of 1/2.
   D) The wave speed increases by a factor of 4.

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5. Which one of the following statements concerning the pressure amplitude of a traveling sound wave in air is **FALSE**?

   A) The pressure amplitude is inversely proportional to the displacement amplitude.
   B) The pressure amplitude is 90° out of phase with the displacement amplitude.
   C) A negative value of the pressure amplitude corresponds to an expansion of the air.
   D) When the displacement amplitude is at its maximum value, the pressure amplitude is zero Pascals.

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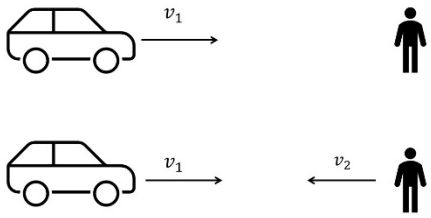
6. A 20.0-cm-long, 10.0-cm-diameter cylinder with a piston at one end contains 1.34 kg of an unknown liquid. Using the piston to compress the length of the liquid by 1.00 mm increases the pressure by 4.15×10^6 Pa. What is the speed of sound in the liquid? [HINT: the volume of a cylinder is given by $V = \pi r^2 L$]

   A) 986 m/s
   B) 1973 m/s
   C) 2206 m/s
   D) 4932 m/s

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7. A stationary person hears frequency f_1 from a source moving toward them at 20.0 m/s. While the source is still moving, the person then moves toward the source with a speed of 30.0 m/s and hears frequency f_2 that differs from f_1 by $f_2 - f_1 = 33.0$ Hz. What is the frequency f of the source? Take the speed of sound to be 340 m/s.

   A) 352 Hz
   B) 568 Hz
   C) 389 Hz
   D) 484 Hz



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8. The simple harmonic motion $y_s(t) = A\cos(\omega t)$ is the result of the superposition of two oscillations $y_1(t) = A\cos(\omega t + \varphi_1)$ and $y_2(t) = A\cos(\omega t + \varphi_2)$ of the same amplitude A. What is the phase difference $\Delta\varphi = |\varphi_2 - \varphi_1|$ between the two initial oscillations?

   A) $\pi/3$
   B) $3\pi/2$
   C) $2\pi/3$
   D) $\pi/4$

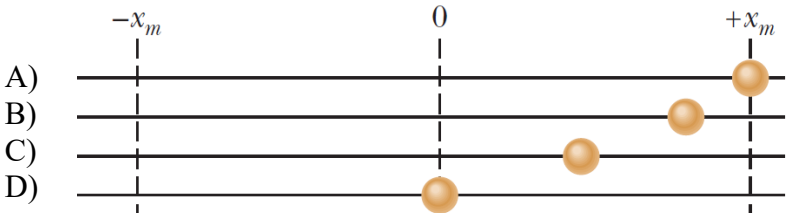
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9. Two ideal pendulums have the same length but one of them has twice the mass of the other one. Which statement below regarding this situation is **TRUE**?

   A) The two pendulums have the same frequency.
   B) The pendulum with more mass has a shorter period.
   C) The pendulum with more mass has a longer period.
   D) The pendulum with the smaller mass has a smaller frequency.

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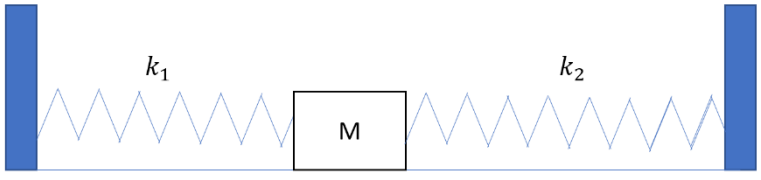
10. Select in which of the following positions of the simple harmonic motion with amplitude x_m the body performing the motion is at rest.



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11. A body with mass of $M = 10.0$ kg is connected by two springs with respective spring constants $k_1 = 30.0$ N/m and $k_2 = 60.0$ N/m, as shown in the picture below. What is the period of the mass's simple harmonic motion?

   A) 3.00 s
   B) 2.09 s
   C) 9.42 s
   D) 15.1 s



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12. The High E string on the guitar has a linear mass density of $\mu_E = 3.09 \times 10^{-3}$ kg/m and is $L = 25.3$ in long. To be in tune, the string's fundamental frequency must be $f = 329.63$ Hz. What tension should be applied to the string? (Hint: 1.00 in $= 2.54$ cm)

A) 34.7 N
B) 850 kN
C) 554.6 N
D) 431.5 N

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13. The sound level at a distance 5.00 m away from a point source of sound is 90.0 dB. If the attenuation coefficient of the air is $\alpha = 1.74 \times 10^{-3}$ m^{-1}, what is the sound level at 100 m?

A) 64.7 dB
B) 25.3 dB
C) 63.3 dB
D) –26.7 dB

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14. An ambulance is passing you on the street. Due to the Doppler effect, the perceived frequency when the ambulance is approaching and when it is moving away have the following relation:

A) They are equal.
B) The perceived sound when the ambulance is approaching has a lower frequency.
C) The perceived sound when the ambulance is moving away has a lower frequency.
D) It depends on the wavelength of the sound when the source is not moving.

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15. You are given two materials. Material one has an impedance similar to that of air and a high attenuation coefficient. Material two has a very high impedance but a very low attenuation coefficient. Starting inside the room and going outward, how would you layer the two materials in order to make the sound insulation of a recording studio? (Hint: the studio must be insulated not only from outside sound but also from sound reflections inside.)

A) Material one, then material two, then the wall
B) Material two, material one, then the wall
C) The wall, then material two, then material one
D) The wall, then material one, then material two

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16. A beam splitter is a device which reflects half and transmits half of the amplitude of an incident light wave. Imagine a beam splitter constructed for sound waves. The impedance of air is $Z_1 = 413.3$ Pa s m^{-1}. What should the impedance of the material of the beam splitter be?

A) 206.65 Pa s m^{-1}
B) 1653.2 Pa s m^{-1}
C) 1239.9 Pa s m^{-1}
D) 137.76 Pa s m^{-1}

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17. You are trying to tune a pair of pan flute pipes, which are air columns open at both ends. One of them is in tune at $f_1 = 261.63$ Hz. The other pipe is slightly too long and when played together they produce a sound with 6 beats per second. If the speed of sound in air is 343 m/s, how much should you cut from the wrong pipe to make it in tune?

A) 1.5 cm
B) 38 cm
C) 0.029 cm
D) 5.0 mm

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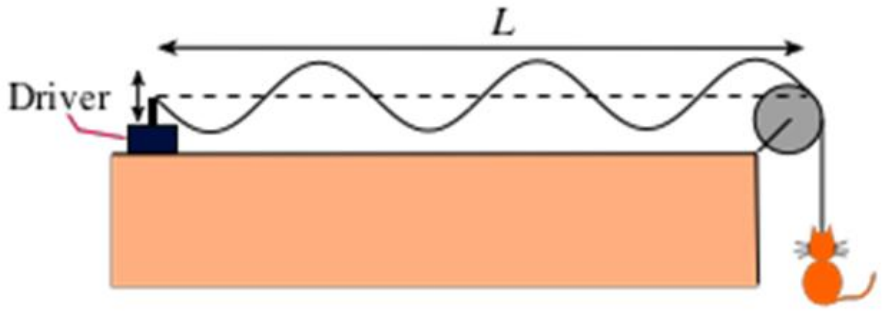
18. A pipe resonates in three successive resonant modes at frequencies 272 Hz, 340 Hz, and 408 Hz. The relevant speed of sound is 340 m/s. Based on this information, select the correct statement:

A) The pipe is open at both ends, and has length 2.50 m
B) The pipe is open at one end, closed at the other, and has length 1.25 m
C) The pipe is open at both ends, and has length 0.40 m
D) The pipe is open at one end, closed at the other, and has length 0.80 m

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Questions 19 and 20 are related to the following scenario:

A quirky scientist places a ceramic sculpture of a cat (mass $M = 3.00$ kg) on the end of a wire that passes over a massless, frictionless pulley, as shown in the figure below. The horizontal part of the wire has length $L = 0.750$ m, and this length of the wire has mass $m = 3.75$ g. A piston moves with simple harmonic motion and drives the left end of the wire such that a harmonic standing wave is created on the horizontal section, as shown in the figure. **Both ends of the string can be treated as if they are fixed.**



19. What is the value of the harmonic index *n* for the wave depicted?

A) 6
B) 7
C) 5
D) 4

20. What is the frequency of the driver in hertz?

A) 307 Hz
B) 205 Hz
C) 256 Hz
D) 358 Hz

PHYS 365 & 369 Formula Sheet

OSCILLATIONS			
$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{L}}$	$T = 2\pi\sqrt{\frac{L}{g}}$
WAVES			
$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \sin(kx \pm \omega t + \phi)$		
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_{vacuuum}}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v\omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v\omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$R^2 + \frac{Z_2}{Z_1}T^2 = 1$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$f = \frac{v}{\lambda} = nf_1$	$\frac{\partial^2 y}{\partial^2 t} = v^2\frac{\partial^2 y}{\partial x^2}$	$\Delta p = -B\frac{\partial s}{\partial x}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$

$x(t) = x_m \cos(\omega t + \Phi)$		$s = s_m \cos(kx - \omega t)$
$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\Phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\Phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$
$\Phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \pm v_S}$
Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2,\ \ldots.$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5,\ \ldots.$
Pipe open at both ends $f = \frac{v}{\lambda} = \frac{nv}{2L},\ n = 1,\ 2,\ 3, \ldots$		Pipe closed at one end and open at the other $f = \frac{v}{\lambda} = \frac{nv}{4L},\ n = 1,\ 3,\ 5, \ldots$

$d\ \sin(\theta) = m\lambda$ for m= 0, 1, 2, … (Maxima- bright fringes)		
$d\ \sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for m= 0, 1, 2, … (minima—dark fringes),		
$I = 4I_0 \cos^2\left(\frac{1}{2}\Phi\right)$	$\Phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$I = I_0 \cos^2(\theta)$

Reflection	Reflection phase shift
Off lower index	0
Off higher index	0.5 wavelength

$a \sin(\theta) = m\lambda$ for m= 1, 2, 3, …. (minima)

CONSTANTS

$c = 3.00 \times 10^8\ \frac{m}{s}$	$R = 8.314\ \frac{J}{k.mol}$	$M_{air} = 28.8 \frac{g}{mol}$	$g = 9.81\ \frac{m}{s}$
$v_{sound\ at\ 20℃ = 343\ \frac{m}{s}}$	$T_K = T_℃ + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	$R = r^2$
$I_0 = 1 \times 10^{-12}\ \frac{W}{m^2}$	$1\ atm = 1.01 \times 10^5 Pa$	$\gamma_{diaatomic} = \frac{7}{5}$	$T = \frac{Z_2}{Z_1} t^2$

MATHEMATICAL FORMULAE

$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$	$\sin(\alpha) \pm \sin(\beta) = 2\sin\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$	$\tilde{A} = Ae^{j\theta} = A(\cos(\theta) + j\sin(\theta))$
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$
$\cos(-\alpha) = \cos(\alpha)$	$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$